

AUDIT DELIVERABLE

STATEFLUX CHANGE-IMPACT ASSESSMENT:

SIM-20260929145158

Empirical, cryptographically grounded evaluation of IPsec configurations, negotiation space frontiers, and deterministic change outcomes.

Generated Date:	29 September 2026
Scope:	40 IPsec Tunnels • 20 Gateway Endpoints
Validation Environment:	strongSwan 5.9.8 • Libreswan 4.12
Report Identifier:	rep-chg-sim-20260929145158-145158

1. Executive Summary

Change Impact Analysis: Proposed policy hardens 10 tunnels, but introduces 7 immediate connection failures and 7 latent rekey failures. Pre-migration remediation is strictly mandatory on blocked tunnels before applying configuration.

2. Current Fleet State & Proposed Change

Target Fleet:	40 IPsec Tunnels • 20 Gateway Endpoints
Change Directive:	FLEET
Enforced Floor:	Mandatory AES-256-GCM • Prohibit 3DES & AES-128-CBC • Min DH 19

3. Blast Radius & Security Delta

HARDENED	UNCHANGED	INCOMPATIBLE	LATENT FAILURE
10	14	7	7

4. Critical Failure Tunnels (Pre-Migration Remediation Required)

Tunnel ID	Classification	Failure Reason / Rekey Window
tn-003	LATENT_FAILURE	Fails at 3600s (1h 0m)s rekey: LATENT FAILURE: Current Security Association remains established and traffic continues. However, at next rekey in 3600s (1h 0m), renegotiation will fail because endpoints will share no common proposals under the modified policy.
tn-004	LATENT_FAILURE	Fails at 3600s (1h 0m)s rekey: LATENT FAILURE: Current Security Association remains established and traffic continues. However, at next rekey in 3600s (1h 0m), renegotiation will fail because endpoints will share no common proposals under the modified policy.
tn-013	LATENT_FAILURE	Fails at 3600s (1h 0m)s rekey: LATENT FAILURE: Current Security Association remains established and traffic continues. However, at next rekey in 3600s (1h 0m), renegotiation will fail because endpoints will share no common proposals under the modified policy.
tn-014	LATENT_FAILURE	Fails at 3600s (1h 0m)s rekey: LATENT FAILURE: Current Security Association remains established and traffic continues. However, at next rekey in 3600s (1h 0m), renegotiation will fail because endpoints will share no common proposals under the modified policy.

5. Phased Migration Plan & Rollback Safeguards

Staged rollout sequence designed to mitigate blast radius and isolate canary risk:

Wave ID	Strategy	Tunnels	Risk
WAVE-01 (Canary)	Branch Pilot	4 tunnels	LOW
WAVE-02 (Core)	Datacenter	36 tunnels	MEDIUM

Rollback Safeguards:

- Full inverse rollback plan synthesized per migration wave.
- Canary stability verification enforced before batch progression.
- Automated pre-rollback state capture validation.